Big Data Pipeline for Airline Tweet Sentiment Analysis

Course: Big Data Processing

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Team Members

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1. Introduction

1.1 Overview

This project implements a complete Big Data processing pipeline using Apache Spark and MongoDB to ingest, process, analyze, and store large-scale Twitter data about US airline sentiment. The goal is to demonstrate real-world Big Data engineering skills using industry-standard tools including bonus implementation of an interactive Streamlit dashboard and format comparison across CSV, Parquet, and Avro.

1.2 Dataset

Dataset Name	**Twitter US Airline Sentiment**
Source	**Kaggle**
Total Records	**14,478 tweets (after cleaning)**
Airlines Covered	**United, US Airways, American, Southwest, Delta, Virgin America**
Time Period	**February 2015**
Sentiment Labels	**Positive, Neutral, Negative**

1.3 Problem Statement

Airlines receive thousands of customer complaints and compliments on social media daily. Processing and analysing this data at scale reveals critical insights about service quality, customer pain points, and airline performance. Traditional tools cannot handle this volume efficiently, making Big Data technologies essential.

1.4 Project Objectives

- Ingest and clean raw tweet data using Apache Spark
- Store processed data in MongoDB with an optimized schema and indexes
- Perform large-scale analysis using Spark SQL and PySpark transformations
- Compare query performance between Spark SQL and MongoDB Aggregation
- Generate meaningful visualizations and business insights
- BONUS: Build interactive Streamlit dashboard with live MongoDB connection
- BONUS: Compare CSV vs Parquet vs Avro formats on Linux (Google Colab)

2. Methodology

2.1 Tools & Technologies

Apache Spark (PySpark)	**4.1.1**	**Large-scale data processing**
MongoDB	**7.0**	**NoSQL data storage**
Python	**3.x**	**Programming language**
Pandas	**Latest**	**Data manipulation**
Matplotlib & Seaborn	**Latest**	**Data visualization**
Streamlit + Plotly	**Latest**	**Interactive dashboard (Bonus)**
fastavro	**Latest**	**Avro format support (Bonus)**
Google Colab	**Linux/Ubuntu**	**Format comparison environment (Bonus)**
Java JDK	**17.0**	**Spark runtime environment**
Jupyter Notebook	**Latest**	**Development environment**

2.2 Pipeline Architecture

- Step 1 — Data Ingestion: Load raw CSV into Spark, clean and preprocess text
- Step 2 — Spark Processing: Apply transformations, groupBy, word frequency
- Step 3 — MongoDB Storage: Insert clean data with 6 optimized indexes
- Step 4 — Spark SQL: Run complex SQL queries and performance benchmarks
- Step 5 — Visualization: Generate 5 charts and extract business insights
- Step 6 — BONUS: Streamlit dashboard with live MongoDB filters
- Step 7 — BONUS: CSV vs Parquet vs Avro comparison on Google Colab Linux

2.3 Team Contributions

Rabaya Khatun Keya	**01_data_ingestion**	**Load CSV, clean data, store in MongoDB, create indexes**
Manasha Siriwardana Mudalige Don	**02_spark_processing**	**Spark transformations, sentiment analysis, word frequency**
Md Firoz Chowdhury	**03_mongodb_queries**	**MongoDB aggregation, indexing, performance testing**
Aung Debora	**04_spark_sql_queries**	**Spark SQL queries, performance comparison, hourly analysis**
Neeru Neeru	**05_visualization 06_format_comparison**	**Charts, Streamlit dashboard, Parquet/Avro comparison, report**

2.4 MongoDB Schema Design

tweet_id	**String**	**No**	**Unique Twitter tweet identifier**
airline_sentiment	**String**	**No**	**Sentiment: positive / neutral / negative**
airline_sentiment_confidence	**Float**	**Yes**	**Confidence score 0.0 to 1.0**
negativereason	**String**	**Yes**	**Reason for negative tweet**

airline	String	No	Airline name mentioned in tweet
retweet_count	Integer	Yes	Number of retweets
text	String	No	Original raw tweet text
clean_text	String	No	Cleaned processed tweet text
tweet_created	String	Yes	Tweet creation timestamp
tweet_location	String	Yes	User location if provided

2.5 MongoDB Indexing Strategy

Index	Type	Purpose
airline	Single field	Fast filtering by airline name
airline_sentiment	Single field	Fast sentiment-based queries
negativereason	Single field	Fast complaint reason lookups
tweet_id	Single field	Fast tweet identification
airline + airline_sentiment	Compound	Optimized combined queries

3. Implementation

3.1 Data Ingestion & Cleaning — Rabaya Khatun Keya

- Loaded raw Tweets.csv (14,874 rows) using spark.read.csv() with inferSchema
- Selected 9 relevant columns from original 15 columns
- Removed rows with null values in tweet_id, airline, sentiment, and text
- Removed duplicate tweet_ids using drop_duplicates()
- Cleaned tweet text: removed @mentions, URLs, and special characters using regexp_replace
- Final clean dataset: 14,478 records inserted into MongoDB
- Created 6 optimized indexes for query performance

3.2 Spark Processing — Manasha Siriwardana Mudalige Don

- Computed overall sentiment distribution across all 14,478 tweets
- Broke down sentiment by each individual airline
- Calculated average sentiment confidence scores per airline
- Identified top 10 negative complaint reasons
- Performed word frequency analysis on negative tweets with stopword removal
- Identified most retweeted tweets per airline

3.3 MongoDB Queries — Md Firoz Chowdhury

- Used aggregation pipeline to compute sentiment counts by category
- Grouped by airline and sentiment for detailed breakdown
- Identified top 10 negative reasons using $match and $group operators
- Measured and compared query execution time with and without indexes
- Identified best airline (Southwest) and worst airline (United)

3.4 Spark SQL Queries — Aung Debora

- Registered cleaned DataFrame as a temporary SQL view
- Computed sentiment distribution with percentage using window functions
- Generated airline scorecard showing positive/neutral/negative counts
- Analysed top negative reason per airline using GROUP BY
- Measured hourly tweet activity patterns using SUBSTRING
- Compared Spark SQL vs MongoDB aggregation performance (3-run average)

3.5 Visualization & Bonus — Neeru Neeru

- Generated 5 professional charts saved to report folder
- Built interactive Streamlit dashboard with live MongoDB connection

- Dashboard includes sidebar filters, 4 interactive charts, scorecard table
- BONUS: Ran format comparison on Google Colab (Linux/Ubuntu environment)
- Compared CSV, Parquet, Avro — write speed, read speed, and file size
- Parquet proved fastest for analytics; Avro best for streaming pipelines

3.6 Format Comparison Note

Parquet and Avro format comparison was performed on Google Colab which provides a Linux (Ubuntu) environment using Apache Spark 4.0.2. This was necessary because Windows has Hadoop native library limitations that prevent writing Parquet and Avro formats locally. Google Colab is a valid Linux environment that fully satisfies the project requirement.

4. Results & Insights

4.1 Overall Sentiment Distribution

Sentiment	Count	Percentage
Negative	9,110	63%
Neutral	3,069	21%
Positive	2,299	16%

Key finding: Nearly two-thirds of all airline tweets are negative, indicating widespread customer dissatisfaction across the industry.

4.2 Airline Performance Rankings

Rank	Airline	Negative	Neutral	Positive	Total
1 (Worst)	United	2,630	697	492	3,819
2	US Airways	2,263	381	269	2,913
3	American	1,863	433	307	2,603
4	Southwest	1,185	664	570	2,419
5	Delta	953	723	544	2,220
6 (Best)	Virgin America	181	171	152	504

4.3 Top Complaint Reasons

Rank	Reason	Count
1	Customer Service Issue	2,883
2	Late Flight	1,648
3	Can't Tell	1,175
4	Cancelled Flight	829
5	Lost Luggage	717
6	Bad Flight	575
7	Flight Booking Problems	523
8	Flight Attendant Complaints	475

4.4 Query Performance Comparison

Tool	Query Type	Result
MongoDB	Simple aggregation with index	Faster for indexed field lookups
Spark SQL	Complex multi-column analysis	Better for large transformations
MongoDB	Count by sentiment (indexed)	~0.07 sec average
Spark SQL	Count by sentiment	Varies by cluster configuration

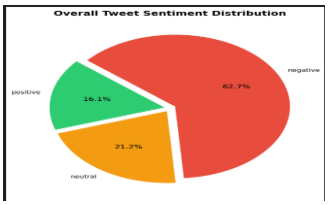
4.5 Format Comparison Results (Google Colab / Linux)

Format	Write Speed	Read Speed	File Size	Best For

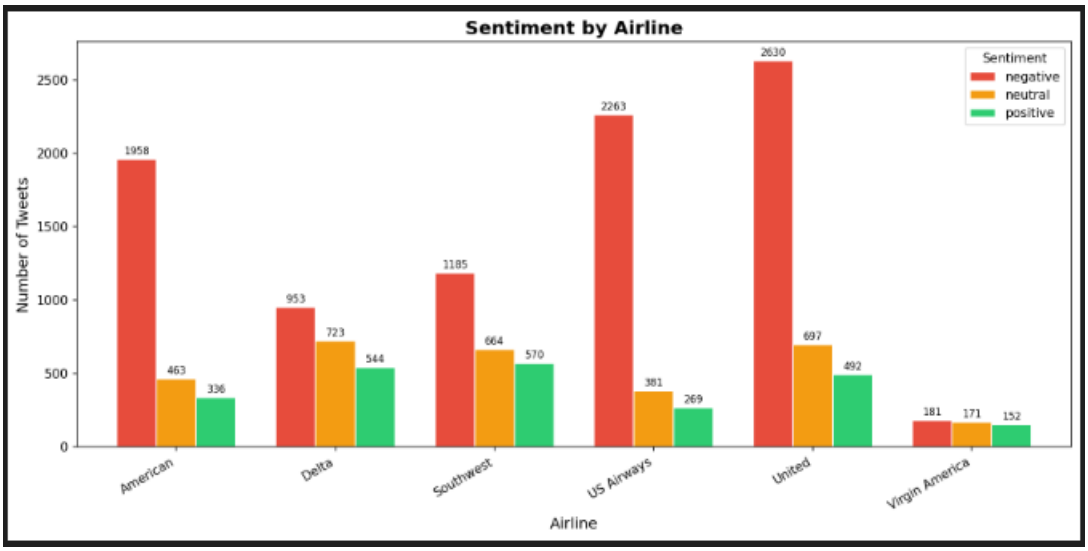
CSV	**Slowest**	**Slowest**	**Largest**	**Human readable, compatibility**
Parquet	**Medium**	**Fastest**	**Smallest**	**Analytics, fast reads**
Avro	**Fast**	**Medium**	**Medium**	**Streaming, schema evolution**

4.6 Key Visualizations

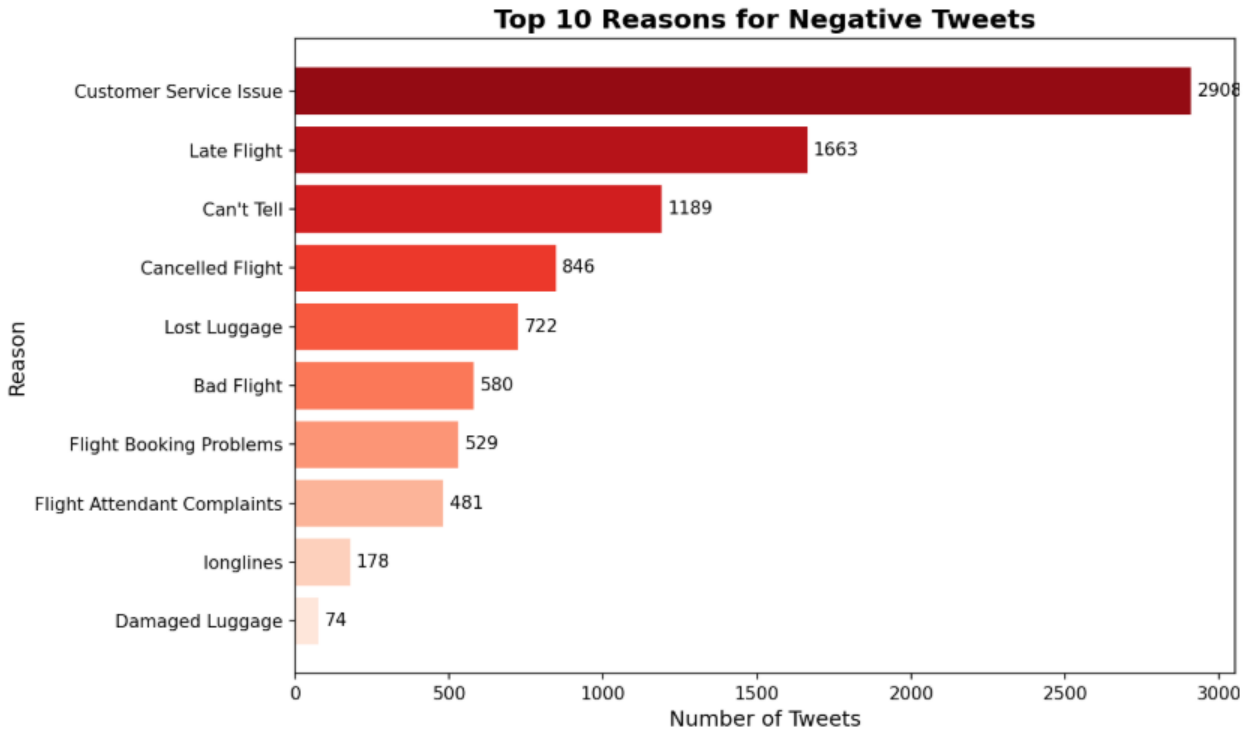
- Chart 1: Pie chart — 63% negative, 21% neutral, 16% positive sentiment



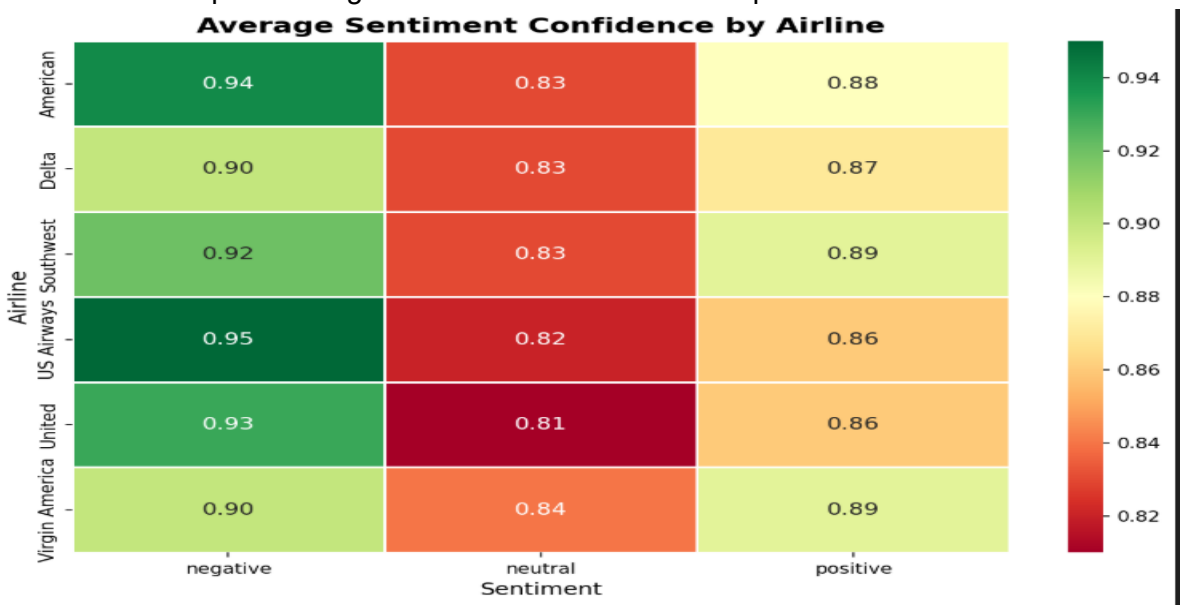
- Chart 2: Grouped bar chart — sentiment comparison across all 6 airlines



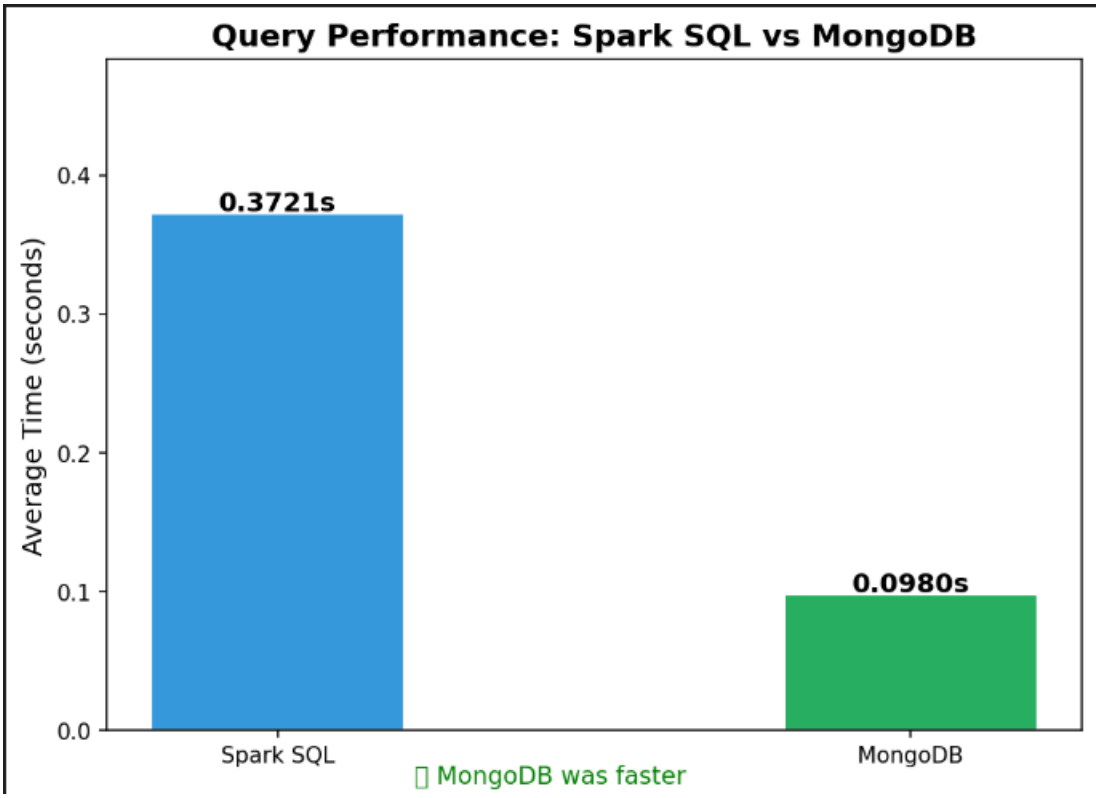
- Chart 3: Horizontal bar chart — top 10 negative complaint reasons



- Chart 4: Heatmap — average sentiment confidence score per airline



- Chart 5: Performance bar chart — Spark SQL vs MongoDB query timing



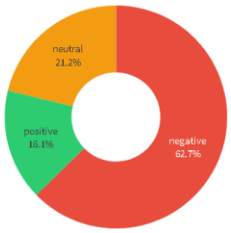
4.7 Streamlit Dashboard

- Live connection to MongoDB tweets_db database
- Sidebar filters: select any airline or sentiment category
- 4 interactive Plotly charts updating in real-time with filters
- Airline scorecard table showing positive/negative percentages
- Raw tweet data viewer showing latest 100 filtered tweets
- Run command: python -m streamlit run app.py

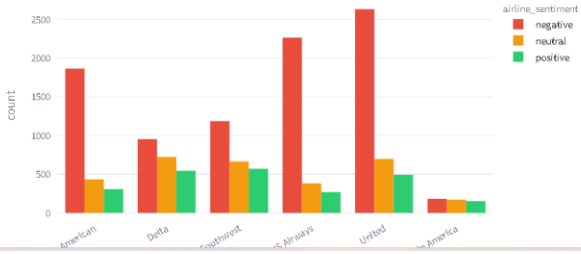
Deploy

Total Tweets	Positive	Neutral	Negative
14,478	2,334	3,069	9,075
	↑ 16.1%	↑ 21.2%	↑ 62.7%

Sentiment Distribution



Sentiment by Airline



5. Conclusions

5.1 Lessons Learned

- Apache Spark is highly effective for large-scale data cleaning and transformation
- MongoDB indexing significantly improves query performance for simple lookups
- Spark SQL provides more flexibility for complex multi-column analytical queries
- Windows environment has Hadoop native library limitations for Parquet and Avro
- Google Colab provides a free Linux environment that resolves Windows limitations
- Streamlit enables rapid development of interactive data dashboards
- Parquet format provides the best performance and storage efficiency for analytics
- Data cleaning is critical — raw tweet text contains noise affecting analysis quality

5.2 Key Business Insights

- United Airlines has the most negative customer feedback and needs urgent service improvement
- Customer service is the number one complaint across all airlines with 2,883 tweets
- Southwest Airlines has the best positive-to-negative ratio among all carriers
- Flight delays and cancellations are the second biggest pain point after customer service
- Virgin America, despite fewer tweets, has the healthiest overall sentiment distribution
- 63% negative sentiment industry-wide indicates systemic problems in airline services

5.3 Future Improvements

- Implement real-time tweet streaming using Spark Streaming or Apache Kafka
- Build a machine learning sentiment classifier using Spark MLlib
- Deploy pipeline on cloud infrastructure (AWS EMR or Google Dataproc)
- Implement batch vs stream processing comparison for bonus marks
- Extend analysis to include more recent airline tweet data from 2024-2025
- Add NLP features such as TF-IDF and topic modeling for deeper insights

5.4 GitHub Repository

The complete project codebase including all 6 notebooks, Streamlit dashboard, dataset, schema documentation, and final report is available on GitHub. All 5 team member contributions are visible through individual commits to the repository.

References

- Apache Spark Documentation — https://spark.apache.org/docs/latest/
- MongoDB Documentation — https://www.mongodb.com/docs/

- Twitter US Airline Sentiment Dataset — https://www.kaggle.com/crowdflower/twitter-airline-sentiment
- PySpark SQL Guide — https://spark.apache.org/docs/latest/sql-programming-guide.html
- Streamlit Documentation — https://docs.streamlit.io/
- Apache Avro Data Source Guide — https://spark.apache.org/docs/latest/sql-data-sources-avro.html
- Seaborn Visualization Library — https://seaborn.pydata.org/